# Artur Iasenovets

**Information Security Research Engineer**


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Research Interests: *privacy-enhancing technologies, private information retrieval, private pattern search, searchable encryption, homomorphic encryption, blockchain privacy.*

## Education

**Chongqing University of Posts and Telecommunications**, Chongqing, China

◇ **M.S.** in Network and Information Security — Sep. 2024–Jul. 2027 *(expected)*

- ◇ Supervisor: Prof. Tang Fei
- ◇ Average mark: 95/100

**Ukhta State Technical University**, Ukhta, Russia

◇ **B.S.** in Information Systems and Technologies — Mar. 2022–Jun. 2024

- ◇ GPA: 4.45/5.00

**St. Petersburg State University of Telecommunications**, St. Petersburg, Russia

◇ **B.S. studies** in Information Systems and Technologies — Sep. 2020–Mar. 2022

## Selected Papers

Artur Iasenovets, Fei Tang, H. Zhu, P. Wang, and L. Liu. "Bringing Private Reads to Hyperledger Fabric via Private Information Retrieval," arXiv:2511.02656, 2025. Manuscript under review.

H. Zhu, F. Tang, W. Qin, J. Shan, P. Wang, and Artur Iasenovets. "EVMK-SSE: Efficient and Verifiable Multi-Keyword Symmetric Searchable Encryption," *Journal of King Saud University – Computer and Information Sciences*, 2025.

Artur Iasenovets and Fei Tang. "MFCTIIF: Multi-Feed Cyber Threat Intelligence Integration Framework," *International Research Journal*, no. 1 (163), 2026.

Piyumi M. S. Rajapaksha*, Artur Iasenovets*, Yinxue Yi, and Fei Tang. "TWAPPA-FRL: Trust-Weighted Privacy-Preserving Aggregation for Federated Reinforcement Learning in Collaborative Intrusion Detection," 2026. Manuscript under review. *Equal contribution.

Artur Iasenovets and Fei Tang. "CBPIR: Content-Based Private Information Retrieval over Structured Databases via Homomorphic Encryption," 2026. Manuscript under review.

## Research Experience

**Graduate Researcher** @ CQUPT — Sep. 2024–Present

- ◇ **HLF-CPIR:** Designed and implemented BGV-based private information retrieval in Go using Lattigo for private world-state queries in Hyperledger Fabric, with query privacy under an HBC peer model based on the IND-CPA security of BGV.
- ◇ **TWAPPA-FRL:** Designed the architecture and implemented CKKS-based private aggregation of DQN model updates using TenSEAL, with IND-CPA-based client-update privacy under an HBC aggregator model and explicit leakage analysis.
- ◇ **EVMK-SSE:** Analyzed the threat model with an HBC data owner and user and a malicious cloud server; reviewed the query-privacy argument in the real/ideal model, based on OT security and the pseudorandomness of CI-rOPRF/PRG under explicitly defined search- and access-pattern leakage.
- ◇ **CBPIR:** Designed and implemented a Rust prototype using OpenFHE, combining CKKS-based FPS/RFPS private pattern search, client-side match-to-record mapping, and BGV-based windowed CPIR for private multi-record retrieval, with explicit leakage and correctness analysis under an HBC server model.

## Awards and Achievements

**Chinese Government Scholarship (CSC) for Graduate Studies** 2024–2027
Full scholarship for the entire M.S. programme at CQUPT.

**Sber Student Accelerator Regional Demo Day** 2023
Completed the Sber Student Accelerator with the AcademAI team and participated in the North-Western District regional demo day.

**Winner, EdTech Project Competition** 2023
Presented LLM-based educational platform and won the competition organized by the MIPT Phystech School, the Fund for Development of Phystech Schools, and Startech.Base.

**Enhanced State Academic Scholarship** 2023
Awarded by USTU for outstanding academic excellence, with the enhanced amount based on documented achievements in the university's index-based student rating system.

**MIPT & Sber Phystech Gigachat Challenge** 2023
Participated in the Phystech Gigachat Challenge Hackathon, developing an LLM-based educational platform using GigaChat and Kandinsky APIs within a 48-hour sprint.

**Kaspersky Cyberimmunity Hackathon 2.0** 2023
Participated in Kaspersky's Cyberimmunity Hackathon in Fall 2023, placing 11th out of 100+ teams for developing traffic monitoring solutions for low-altitude airspace security.

**First Degree Diploma, Severgeoecotech** 2022
Awarded first place in the Modern Information Technologies section of the 23rd International Youth Scientific Conference.

Supporting certificates

## Industry Internships

**Information Security Engineer** @ Positive Technologies Aug. 2024–Feb. 2025

- Developed Python tools with unit-test coverage for validating Windows paths, network configurations, and endpoint-control policies.
- Built an automated malware-processing pipeline with an Airflow DAG that collected samples from MalwareBazaar, stored artifacts in S3, and scanned them with YARA.
- Analyzed EVTX logs, PowerShell, and malicious VBA macros; extracted IoCs and mapped TTPs to MITRE ATT&CK for CVE-2020-5902 and CVE-2023-38831.

## Other Experience

**Software Engineer** @ AcademAI Sep. 2023–Aug. 2024

- Developed the application layer with Next.js/TypeScript application, Python/Flask services, PostgreSQL/Prisma data layer, and Auth.js authentication for an LLM-based educational platform used by computer science university faculty and students.
- Integrated LangChain-based generation workflows with AsyncIO and Redis for asynchronous LLM orchestration and intermediate state management.
- Containerized the application with Docker, configured Nginx, deployed to cloud-hosted VPS infrastructure, and built the GitHub Actions CI/CD workflow.

**Junior Software Engineer** @ ZVEK "Progress" LLC May. 2022–Aug. 2023

- Designed and developed a full-stack warehouse QR-inventory system using Python, FastAPI, and MySQL, replacing a manual tracking process entirely.
- Supported approximately 1,000 records and 20+ concurrent users; added unit-test coverage, Git-based development, debugging, and Linux deployment.

## Skills

- **Programming:** Go, Python, Rust, TypeScript, SQL, Bash, LaTeX
- **Systems and infrastructure:** Linux, Proxmox VE, VMware, Docker, Git, CI/CD, Nginx, Ansible, Airflow, S3
- **Cryptographic libraries:** Lattigo, SEAL, TenSEAL, OpenFHE
- **Cryptography:** Lattice-based, symmetric-key, public-key
- **Security engineering:** YARA, malware analysis (PS/VBA), IoC/TTP analysis, MITRE ATT&CK
- **Languages:** Russian — native; English — C1; Chinese — HSK3